

Active Inference BookStream #001.02 ~ "Governing Continuous Transform

BookStream | 1:19:04

Hello everyone, it's December 13th, 2022.

We're in ACTIV bookstream number 1.02 on the book Governing Continuous Transformation.

Off to blue.

Hi everyone, we are the ACTIV Inference Institute, a participatory online learning institute that is learning, practicing, and thinking about applied active inference.

You can reach us here at all of our media links.

This is a recorded and archived live stream, so please provide us with feedback so we can improve upon our work.

All backgrounds and perspectives are welcome here, and we will be following good video etiquette for live streams, hopefully.

Get in touch if you want to participate, either in this bookstream or the .zero contextualization streams or in another live stream or maybe in the discussion stream.

You can reach us always at active.org or by email.

Okay, so here we are, bookstream 1.02.

The sections will go through the book map and the roadmap, the keywords and the claims, and we'll be continuing the discussion of Governing Continuous Transformation.

So I don't know if we want to just maybe mention a couple of things, one or two things that we're excited to discuss more today or learn more about from this section of the book.

Tyler, is there anything that stood out for you specifically?

Or maybe we want to give an introduction and then maybe say one or two words about the book, the chapter?

Yeah, sure.

So I'm Tyler.

I work more on the applied side of thinking about governance, organizational design, both work in like the DAO tooling space and also protocol design.

I also work in a lot of like corporate settings or as a strategy consultant.

I think for today, what I'm excited about is that, you know, I went to business school, something that I hear a lot about, like some of these, you know, organizational design frameworks that Bajan references in the book.

It's really interesting to compare that directly to free energy governance and compare and contrast, see where the gaps are.

So I'm excited about exploring that more today.

What about you, Blue?

I am really excited to kind of link.

And so I'll do this right up in the very beginning, but to link what we have talked about

in the Active Inference Institute over the last couple of years about biological organization and hierarchical like information flow in biological systems, I'm excited to connect that to how that works in an organization.

So, so I just, as I was reading the book, that was something that stood out to me as something that's maybe necessary for like a more elaborate discussion of the text.

Cool.

Daniel, anything?

All of the comments at the roadmap, but I'm Daniel.

I'm a researcher in California and just looking forward to this discussion.

Cool.

Thanks.

Okay.

So here we are at the book map.

We are in chapter two, sensing sense making and strategic renewal.

And the roadmap for today is just two sections.

These were structured actually by the author, strategic cognition, sensing and sense making.

And the second section is strategic renewal.

Did you want to make a comment here, Daniel?

At the roadmap?

At the roadmap?

At the roadmap?

Not yet.

Thank you.

Okay.

Okay.

So lots of keywords in today's discussion.

So we'll just jump right in and hopefully we put them in an order that aligns with the order in the book and included some commentary and claims from the text as well.

All right.

So Tyler, do you want to read this first part of the abstract?

Sure.

Since the sense making form one unitary acts and together consider the non operational cognitive medicability or dynamic capability.

There's no sensing without sense making and vice versa.

And there's no organizational sense making in the absence of a clearly defined deep existential corporate purpose.

Generally, the processes of top down bottom ups hypothesis testing prediction error minimization.

Rather than top down jurisdiction or guiding the path navigating a discontinuous world.

There's neuroscience is predictive coding that allows us to take management researches strategic cognition and more specifically, the net is a structure and cognition and one step further to provide a more complete and practice relevant framework for applying sensing and sense making in the context of strategic renewal.

We'll take next part.

Sure.

Thanks.

Daniel, do you want to read this part?

No.

Okay, I'll do it.

Effectively, sensing and sense making are empowered by the circular causality of top down knowledge structures, prediction models, predictions, hypotheses, and guesses.

And bottom up stimuli data stemming from resource markets.

By default, uncertainty and complexity are subjective concepts but are consistently deployed in management research as objectively determined.

Environmental sensing and sense making are nonsensical in the absence of having set the firm's purpose and context in the first place.

Once we define sensing, sense making and strategic renewal in form of action centric circular causality linking purpose, prediction models, predictions, and prediction error minimization, a new logic of strategic renewal emerges dynamically connecting structure hierarchy cognition and capabilities.

And so that I just really wanted to bring this in here.

We're talking here about top down and bottom up and I know that that's like there's management styles and structures that are associated with top down and bottom up like management styles.

But I really wanted to bring up this paper, the information theory of individuality by David Krakauer and colleagues at the Santa Fe Institute.

And I bring this up because really they try to define what is an individual.

So even in a biological system, an individual like we are, we have more or just as many cells inside of our body that are not human as we do cells that are human.

Right. But like if you were to take all of those non-human cells out of my body, like I would not be the same person.

Like I would not function optimally at all.

So I'm more maybe than just what is my DNA.

Like there's a, I'm a colony, right?

Not, not necessarily like I'm one organism.

I'm many organisms to function appropriately.

And so really these guys tried to define how you like contain an individual, like what is worthy of this Markov blanket?

Like we talk about the Markov blanket as the boundary between internal and external, but like, where do you

put that?

Am I drawing it around like my human cells between my human cells and my non-human cells?

Like that doesn't really make sense.

Right. And so they talk in this paper a lot about, and they do it in a hierarchical way, which is why this, the biological levels of organization are here because you can define an individual like at the level of a cell or a tissue or an organ, organism, population, community, and so forth.

And this even applies to organizations like a company.

So, so here, what is critical for, to be defined as an individual is this bi-directional information flow.

And, and I think that this paper is very elegant and great.

So you have to have information flowing top down, and you also have to have information flowing bottom up to be like worthy of a Markov blanket and considered an individual in this paper.

And so I really just kind of wanted to bring that in because they talk about it.

Bijan talks a lot about how you need this bottom up and top down, and they have to work together.

And he talks about that a lot in the, in the book for a firm to persist.

And so like really it's necessary for anything to persist.

Like you have to have that bi-directional information flow to earn your Markov blanket, so to speak.

So I just wanted to kind of bring that up here before we jump in to the additional context talking about top down and bottom up in the book.

Did you guys have any comments here?

Daniel?

Daniel K. Yeah.

What I was going to suggest that the roadmap was a quote from the abstract subjective concepts pragmatically applied, which I think deals with a lot of informational concepts.

And then like you brought up, if you define self spatially or genetically, you're going to get some boundaries, which may be relevant in certain situations, but also might give a misleading or partial view of the total system.

And then the challenge is when we move to this informational or statistical theory of nested selves, then organizations and their niches can have many different looking causes and components.

So it's very challenging and it's not the org chart.

It's not the computer access chart.

It's something that's somehow different.

And I think that'll be relevant to pull out what is being described and what should people's consideration be?

What do we actually gain by saying that systems are nested, which isn't a novel insight?

Yeah, that's great.

And then like also, you know, and talking about that, like, you know, bi-directional information flow like the there's a certain amount of salience that has to be there also like information can be flowing both directions.

But if the information is irrelevant, then that's going to, you know, not not be the cause for perseverance.

Right.

Okay, cool.

Okay, so here we are.

Strategic cognition, sensing and sense making and just early quote from this chapter.

Do you want to read this, Tyler?

Is that okay?

All right.

Strategy, initiation, and formulation, defined as a complex activity consisting of scanning, sense making, and strategic decision making are at the heart of the firm's collective interpretive power, emphasizing its core capabilities, timely and accurate

breathing of expected environmental change in the corresponding need for resource reconfiguration and pursuit of adaptation.

Cool, thank you.

Okay, so strategic cognition, they used the definition presented in this paper, the cognitive perspective in strategy and integrated review and they reference this paper quite a bit.

So I just pulled out this diagram of strategic cognition and pulled up the definition that was used in this paper.

So they say strategic cognition is discussed in terms of its constituents, structure and process.

By structure, we refer to the relatively stable characteristics and repeated patterns of behavior, and by process to the cognitive activities taking place in organizations.

Just a note on that, the focus in this book is on kind of a certain type of corporation with a board of directors, for example, which is an important category.

So this, I'm also just imagining someone who is navigating spatially as part of like working in a taxi service.

Also, strategic cognition is coming into play, though the wording might be framed a little differently.

But it's similar questions about sense making and just different implementations of different kinds of organizations.

And I wonder how, like strategy in terms of this kind of strategic cognition is related to like the systems engineering approach.

Is it like aligned with that or maybe not?

That's something be interesting to, to know more about.

Do you have any comments here, Tyler?

Yeah, that's something that to build on Daniel's point, like how specific what Bajan is talking about is to boards specifically versus other organizational structures.

And just like a comment on that, my initial feelings on so far where we are in the book is that it's really not.

I'm not seeing why we need to reduce the scope to just like organizations that have boards, especially since like most companies I've been in a lot of the important decision making is really happening at the level of really more of the executives in the C-suite.

And Bajan might argue like, hey, that's actually a problem.

It should be more coming from an active board.

But it's kind of interesting to see like where are the limits of what Bajan is talking about and how specific is it to a specific kind of organizational structure?

Or another angle is in the body, the decisions are being made in a distributed nested way.

And so that kind of organization, I think we'll read on to find, will have resonance with this.

However, who should the strategic communication and cutting edge research on the strategy governance upstream decision making?

There's a very certain audience.

I think it makes sense to communicate that to.

And also this is the threads being linked for a first time in a dissertation style book.

And I could imagine different kinds of presentations and implementations being presented differently, not necessarily like an academic published dissertation.

And that leads me to wonder, perhaps strategic cognition can take place at many levels in a hierarchical system.

Like maybe, you know, an individual like a project or a team is doing strategic cognition on the execution of their goals and tasks.

And that happens, you know, at the level of the individual and the team and the project and the, you know, department and then the firm and then the board.

So perhaps it occurs in several places.

So like when I've been reading board, when the Bajonis is board, I read that as like counsel.

And that's my like interpretation of that.

And that can happen at many levels as well.

I mean, like when you say counsel, I think attorney, like why do we have lawyers involved?

I mean, counsel is in like high counsel, Supreme counsel.

Okay.

Okay.

All right.

Awesome.

Okay.

So sensing, I pulled this out to Tyler's, you know, protests.

I wanted to just maybe talk a little bit about sensing and sense making and talk a little bit about them separately, but but they are sensing and sensing, sensing and sensing and sense making form one unitary act and together can be considered a non operational cognitive meta capability or dynamic capability.

And this reminds me of the time that I asked Carl about like when I'm sensing, what am I sensing?

Like if I am I sensing, like when I look at a red balloon, am I sensing red balloon or am I sensing like 700 nanometer wavelength of light and like round oval shape?

Like what is actually being perceived via the senses?

And so like there is actually, and Carl answered me, he said red balloon, like go over the sun.

Like go away from the red light and go more toward the red balloon.

And so like that, that answer really says like, okay, there actually is not sensing without sense making like they are inextricably linked because if I'm actually sensing red balloon, like then I'm actually like I'm going beyond what like my eyes see or like what my nose smells or whatever.

Okay, and effectively sensing and sense making is a form of organizational learning action centric hypothesis

testing that connects the individual level with that of the firm.

Do you guys have any comments here? We can move on to sense making or stay in sense making maybe.

All right, Tyler, this is yours.

All right, sense making. So purpose goal directiveness. So the world's fundamental, the world's models underlying prediction models fundamental to sensing and sense making.

So dissonance is a function that a preset goal generating predictions and submitting those predictions to continue to take continuous error minimization.

Since making models are constantly adjusted through error and deviation querying feedback loops, coupling top down predictions and bottom up stimuli and circular causality.

The sole objective and minimizing prediction errors and surprise.

I need free energy. Fundamentally sense making is about designing, you know, the same systems.

And so the quote that I really liked from the paper that is actually from wake a much fun, that right is sense making is about how people deal with situations where there are too many or too few meanings.

Like if you're in a startup and you're trying to figure out what you do next, like your strategic directions are like kind of infinite because you haven't really built that many.

So like there's like all these things directions you can go things you're hearing and ecosystem like, oh, if we did this, look, we did this and like that all could be meaningful.

But at the same time, it all collapses into like, what are we actually doing?

Like, are we even building anything right now? Like, what's the point of any of this?

And so a lot of how that kind of shows up in your like, for example, in a health context, that was showing up in the body is kind of like illness or the feeling of agitation, anxiety.

And you kind of have that same embodied feeling both in yourself and in the organization of like, hey, like what we're doing isn't working.

We're having all of these like communication kind of issues where like the breakdowns where I'm not, you're not receiving what I'm telling you that the information I'm telling you about problem isn't being received correctly and acted upon.

And it just like boils up to becoming a bigger issue to where something needs to happen and some action needs to be taken.

And so it's really like this social process, not something that's like the CEO of me like, wow, like, let me like go in here and like figure out the root issue like a plumber would and then like tell you tell the organization what happened.

But it's more of this kind of like feeling of agitation that happens throughout the organization and through the social process being like, what is going on?

What are we doing? People getting increasingly flustered, something is birthed in that process.

And something that actually had like a personal realization when I was reading this chapter because I used to work as a strategy consultant and strategy consultant, like kind of realized about that job is that it's purely about sense making.

So a lot of times people think of strategy consultants as like, oh, you're doing like a complicated quantitative analysis.

You give that to the board of directors, they make a decision super top down.

But actually where we spent most of our time on was socialization, like kind of talking to people, like kind of knowing when we went into the organization, like roughly what was going on in a certain way.

We collect data, I kept talking to people, but even when we were like, we're pretty sure we had the answer.

Most of the job was actually framing and communication and like talking to a bunch of people and using the language that they use and making sure that there is a shared narrative between everyone.

And so like that was where the real work is.

And I was like, oh, that's interesting.

Like management consultants are professional sense makers.

And I think a lot of times a lot of organizations are actually embedding management consultants within the organization as a way of like sense making as like a capability that an organization would have.

And so like an example of something I had as a management consultant where you kind of feel this dissonance was like I worked for a lot of like health insurance companies.

And so health insurance companies were having this problem where, well, at least in the US skyrocketing health care costs.

And if you're a health insurance company, you're basically a payment processor.

And so you don't really have any levers to fix that.

And so there's this issue of like, okay, well, we, we need to rain costs, but like we don't have any levers to do that.

And so what that means is that you have to become more of a like a care manager to some extent.

And like that is a totally different way of like contextualizing like who you are, both on the organization level and on the individual level of like, what is my purpose here in this role in this organization?

What is this organization trying to do and getting to the care management side, like totally needs to shift that.

And so that shows up in the organization.

That's a lot of dissonance, a lot of angst, and also like a lot of pain.

And it ends up meaning that both the organization needs to change, but also this individual needs to change themselves and how they relate to that organization.

And so that's why I like this cartoon of like, I've been restructured as like a lot of times it's like you are have to recontextualize yourself while also the organization is recontextualizing like what it's trying to do.

So, yeah, I'll leave it there, Daniel and Luis, you have anything to add on the slide?

Well, that's very insightful.

and it reminds me of the two ways that the entities can reduce free energy by changing their mind and changing the world through action and this is an approach where we can describe that process happening at multiple nested and interacted levels and even if the niche changed around one the world would be different the regime of attention would be different the action perception loop would be different and that would at the very least induce anxiety and uncertainty cognitively so it's really interesting what you added and i just love your cartoon that you pulled out it's good okay uh so here is a claim from the book from the chapter on a more operational level sense making's very effectiveness is shaped by three critical dimensions first the clarity of a

timeless corporate purpose second a dissonance centric quest for and optimization of filtering anomalies as anomalies become shared sensibleness should become stronger and see an an activist worldview i.e environment is not exogenously predetermined but is enacted through the social and action-centric construction of shared meaning um and so like let's talk about dissonance a little bit um going back to this quote unless you guys have comments here no okay so fundamentally firm level sensing and sense making grounded in dissonance theory are a social process rather than a singular activity dedicated to detecting anomalies and i think this goes back to what tyler was actually just talking about um but i looked up like the theory and this is the the citation um that bijan used and in this text um festinger says in short i'm proposing the dissonance that is the existence of non-fitting relations among cognitions is a motivating factor in its own right by the term cognition here and in the remainder of the book i mean any knowledge opinion or belief about the environment oneself or one's behavior

cognitive dissonance can be seen as an antecedent condition which leads to activity activity oriented toward dissonance reduction just as hunger leads to activity oriented toward hunger reduction it is a very different motivation from what psychologists are used to dealing with but as we shall see nonetheless powerful

um and so when we talk about sensing and sense making being grounded in dissonance theory this was something

tyler and i uh had discussed um offline back channel uh we were we were talking about dissonance theory and and what

we're interpreting this to mean here is that um it is this like disjoint activity like like the like a mismatch between maybe like individual and firm level cognition or firm level cognition and like societal expectations or anyway like there's a gap um in that causes like a kind of discomfort that needs to then be filled um but i would be uh excited to hear maybe bijan talk more about how it's grounded in dissonance theory like um just because it's it's it was kind of unclear from the text like what exactly that might mean um but if you guys have any comments here daniel yeah um in active inference we might think

about dissonance as being related to a divergence like a kl divergence and so dissonance isn't a state of exception and kind of a binary switch there's always some non-zero level of divergence between expectations and incoming sensory data if only because the generative model is coarse grained so in the decimal points the world always is going to be somewhat surprising or at least evaluatable in terms of surprise so again instead of centering for example absolute value attainment and then seeing uncertainty as a secondary characteristic we use divergence based statistics to incorporate this idea of dissonance or divergence driven behavior and put that at the center of the imperative for action

yeah i i interpreted uh bijan's bring a dissonance theory as like way of bridging but the individual cognition to like the social cognition and like how the feeling of anxiety manifest in organization and so like i think dissonance theory at least in this quote might be referring to the individual specifically um and have that individual but anxiety of like hey i thought i was going to get a promotion this year i didn't get a promotion this year something is wrong or like hey we i worked

really hard on this feature it didn't increase conversions in the way i thought it would that feels bad that's not motivating for me right that's like on the individual level but then you can there's another way you can interpret that dissonance on a more of like a social level as a way and like more of a social cognition cool all right non-uniformity tyler all right um all right it's critical that the environment throughput listening to what wants to emerge is neither logic nor goal trapped but open to non-uniformity in fact the throughput of non-uniformity is a measure of firm survivability uncertainty and surprise are not to be avoided but to be minimized through action hypothesis testing centric engagement with the world as the underlying world model consequently and generally really gains an informational edge and so we talked about this a little bit uh during the last book stream but a lot of leaders want to imagine that they're like slowly ironing out non-uniformity uh and that like they're you're creating this perfect perfect organizational design and it's actually kind of funny like there's this trap that a lot of leaders get into and i've actually gone into this myself where like it's kind of like utopia design almost actually it's exactly the same thing as utopia design where you're like i'm gonna have this model of like where i'm gonna like all these problems i perceive in the world like i'm just gonna fix by just like top down just like god mode figuring it out and then like everything is gonna be there and like we're not gonna have to worry about anything anymore so we create um and then like invariably like you're two weeks you create a new org design two weeks and it works for like a week and people are excited about it and then like you start seeing kind of ruffles at the edge you start like trying to patch things up and then it like all disintegrates um and then like i think a lot of leaders are like oh well like you know people we have the wrong people people are you know following process like i want them to right and it gets really frustrating um but i think even when it goes well it's just like you were missing out on kind of like how generative that non-uniformity can be uh and where basically like if people are too rigid in their process you're you're not carving out space for anything new um and so like the way that he talks about non-uniformity in this case is that you minimize it through kind of action-oriented uh hypothesis testing you're not like wallowing in this uncertainty there's a way that you can capture that and take ownership over uncertainty and use it for something that's really like beautiful and productive um and so like for example you know i've had jobs where i was a product analyst and i worked for this like consumer logistics company where we like go into people's homes and like pack up all of their things so it's like a very very complicated job right and so we would have we would like figure things out like okay wow our um times for this kind of job in the city are going up a lot and we would see that in the data and we're like wow this is really inefficient and that we would like that manifest itself as like some kind of anxiety or dissonance and then we would like start um we do analysis and we'd be like oh it's this kind of job is this kind of job with this kind of furniture then you like talk to people to talk to some of these movers and you find out oh they don't have the right equipment on this uh they don't have like this kind of wrench on these particular jobs they're like that's interesting then you socialize that that information figure out from operations that that wrench isn't consistently being returned and they're actually

losing a lot of uh their equipment to be able to do that job then you find out like uh from other operational leaders that they're not being funded appropriately they don't have like discretionary spending to be able to like actually pay for those kinds of that kind of equipment right so it's like through that that anxiety you kind of like are action oriented and can figure out like hey like this is not something that the situation isn't totally screwed like we can actually keep moving forward and like testing figuring out like what's going on and taking action accordingly that makes me think of pain and soreness and redness in the body like signals have to propagate in order for things to change and that could be in an allostatic or an anticipatory or in a reactive or in a traumatized way from the organism's perspective but the only way that we're going to find out about it is through this kind of like active perception individuals or nested levels above them making an action-oriented decision to like sense out different information and control attention which is actually seen as a policy selection over cognitive states

sorry my mouse was sticky um i am also reminded of biological systems but maybe for a different reason so in biology um non-uniformity is critical to existence like you have a system at equilibrium in a biological system that's death um like so so this non-uniformity and i think um he even brings it up like like as critical i don't know if i quoted it here but he brings it up as critical to um you know survival you have to have non-uniformity if everything's the same it's like dull boards like it's it's just it's it's so um important to have variety um variety is the spice of life right okay cool all right purpose so purpose and goal directedness i.e the world model's underlying prediction model is fundamental to sensing and sense making and like this is the world model's underlying model so i just wanted to point that out that there's some some nestedness going on here dissonance is a function of a preset goal generating predictions and submitting those predictions to continuous error minimization environmental sensing and sense making are nonsensical in the absence of having set the firm's existential purpose in the first place purpose not strategy is the reason an organization exists its definition and articulation must be top management's first responsibility um and this reminds me of our like management um organization in the active inference institute like what is the purpose of the institute um and and just trying to kind of outline and highlight these key things like like we can't have a strategy without having a goal to like like what is that strategy trying to achieve what is the point right um if you guys have comments here too that'd be awesome just that uh is this meant to be a sentence in one language or another language even in the same language or a formal language it will mean different things to different people it'll be associated with different envelopes and portfolios of risk and positive or negative consequences also deemed positive and negative by a different worldview and contingent upon all these unknowns unknowns so i think amidst that is where we can explore a lot of uh interesting experiences i'm sure and what has worked and what hasn't worked in the applied settings and then also start to synthesize it with the research on cognitive science and organizational cognition and that's i think what this book is about like moving beyond just recognizing that complexity and starting to apply active inference and propose free energy governance as an approach to step in here and be useful as well as grounded

yeah just to add on that like i think i'm curious if john ever gets to a point in the book where he really gives some guidelines for thinking through how to design purpose more explicitly um because generally this is done extremely poorly even in like major corporations where they're like our purpose is like bringing smiles to our consumers or something like something just like super vague and big use that like actually doesn't really help you at all it's more of actually like a branding exercise um and more like a corporate social responsibility maybe not really anything about like purpose and when purpose is done well it's really more like i'm you guys might disagree with this this definition but like i think it's kind of more like a heuristic it's like a decision rule of like we do this and not this uh and it's a way of collapsing information and just being like we let's not think through everything we could possibly be doing right now like we already decided this this is what we're doing and it's a way to not think about things uh too and so i found this to be especially important in like decentralized context as well um and i think this is kind of what free energy governance is getting

towards like where you have more bottoms up uh stimuli that you're processing and changing like these top-down models and it's really important in that context because you can't be managing everybody like really intensely like this is exactly what you have to do but rather you have to give a decision rule that a decentralized group of people can use independently um so especially as we get into more like a free energy governance kind of world like purpose is super important but like daniel touched on like it's not always totally clear what like a good purpose even looks like and i hope john has a very crisp answer later on for what good purpose is so like here i am like the non-business person um and tyler you said you don't read like business books but like i've read more than i like ever wanted to read um it just in like my own little startup side venture um but but i read uh jim collins good to great book and he really uh talks about like purpose is really as defining what you do and what you don't do and and he really i mean it kind of affronted me i like he talks about the fox versus the hedgehog like you know that the fox um does lots of things very well he's very like clever and can do many things and the hedgehog like only has one thing they do one thing like that's it they roll up into a ball right like that's all a hedgehog ever does and so like in um this fox hedgehog analogy like as a company he he says you want to be a hedgehog and i'm like but but i want to do so many different things but but really that having especially like when i think about a decentralized thing like we do one thing like i don't know he talked about like many different companies but but like you know wells fargo as a bank was a good example like we are creating the best bank in the united states of america we're never banking outside and and like maybe it started regionally and expanded a little bit or even southwest we are only going to fly in the southwest like they just defined one thing that they were good at that they could just be better than everybody else at and just ran with that and and that is kind of like one of the defining factors of a great company in his view yeah exactly to build on that a little bit like it's been a long time since i read good to grade but jim collins is kind of like the prototypical like top down or what some would call like orange style of um management where it's very achievement oriented it's very much about like the ceo as being the hero uh forcing

their way and like telling the organization what to do and focusing everybody really intensely and so when you talk about the fox and the hedgehog like it's interesting he says you want to be the hedgehog because usually when i hear people talk about the fox and the hedgehog it's like you don't want to be the hedgehog like forcing everything into your narrow world model um and so i think when he says be the hedgehog it's really coming from this top-down perspective and it's really not getting into like the free energy governance model of like we're like ideally you want to be both the fox and the hedgehog we're like you really want to have um a unified world model that you're like hey this world model is like not true by the way like we all know that this is subject to change and we're going to be like actively changing this but like we all have to be on the same page and so we're going to have this somewhat arbitrary belief of what our purpose is that we're going to continuously evolve over time in a fox-like manner and so it's like that's a really hard thing to communicate uh and it's like very subtle and i've tried to communicate this in the past often ineffectively um and so yeah i'm really curious if there's like ways to you know better combine that fox and hedgehog you know model so like totally now we're spinning out on a tangent but i just have to say like i don't know like i get the whole fox thing like i i am very much like aligned mentally with with how like a fox type of behavior like because i'm just like oh squirrel like like let me go chase that but but um the hedgehog thing like i was thinking like okay do you really want to be a fox and a hedgehog or do you want to be like multiple hedgehogs like i had this like vision of like dropping water on the hedgehog like the little gremlins you know and you drop the water and then they multiply and then you have like you know a little field of hedgehogs that are all like doing many different things in a very great way but all siloed and you know they're one little thing okay we have to we have to go on or we can talk about gremlins all day long uh well so for heuristics i mean this actually fits into purpose because you could think of purpose like as a heuristic and i'm not sure john would agree with that but that's how i think about it um and so reed at least a few his definition was you know cognitive heuristics are a form of cognitive simplification or rules of thumb and tools that empower the human mind what the poet john keiths called negative capability the ability to survive and thrive uh and in uncertainty um and so yeah there's just a way of like collapsing information overload and finding a way forward despite like not having the ability to process everything at the same time um but then you know bajon goes on later to say that heuristics are fundamentally anchored in the world view of environmental adaptation rather than enactment so i.e resolving uncertainty is rather treated as a matter of adaptation to an exogenously determined game this book rejects environmental adaptation as a viable survival strategy in a perpetually discontinuous one-shot interactions uh environment um and so to just like illustrate this a little bit so that you know obviously i think we could all understand like a heuristic may not be very good and you could use the wrong here a second have you know but just like to make this more concrete about like how heuristics can be confusing is that here are like two like really common heuristics for um financial planning especially in the context of startups so one is called default alive and this was from um paul graham he's the founder of like combinator and so he says okay a company should cut costs

such that if you assume constant revenue growth the company is going to make it a profitability with the money it has left and so like that totally makes sense especially in a startup context where like extreme uncertainty but then there's like another decision rule which is more about expected value so a company should choose the financial decision with the highest expected value expected payoff of outcome $\times$ $u(m)$ or outcome times the likelihood of the outcome and so these aren't like necessarily incompatible $u(m)$ well like they can be complementary to each other but they could tell you to do very different things uh especially if like an expected value of like hey we can crater the business right now is like a 10 likelihood but we can also have a 90 likelihood but 10 likelihood we make a you know a hundred billion dollar company right $u(m)$ and so like that would lead you to do a different thing than default alive and so a lot of times business leaders are like juggling both of these at the same time and not necessarily like realizing that they are contradictory heuristics and that they are there are limits their heuristics $u(m)$ so yeah i'll leave it there i think actually a couple other things that he says to you are that like you know simple heuristics can outperform more complicated ones as well and it's like actually not even clear on an organizational level $u(m)$ how effective heuristics are because a lot of times when they do studies on heuristics is with individuals where $u(m)$ everything is there are not other intelligent actors who are making anticipations about your strategy whereas in like a a business setting there are and so to take like a very simple example of this like with design there's for a long time was this movement of like the heuristic for design it's like oh everything needs to be super legible and clean $u(m)$ and like you would use like helvetica 16 point font like really like we want this to be clear and we want information hierarchy and like that matters far and above everything else but then the problem is that everyone was doing that and it got really monotonous and so the actual optimal strategy in that case was to design something totally wacky and insane and so over the last three years there was like this especially with the web 3 just like absurd like looked like they were like geocities pages from the 90s where it was like looked awful and it was like not accessible and it was gross but it was really interesting to look at $u(m)$ and they were using fonts that were totally unreadable but it was interesting and it was this competitive situation where like the obvious heuristic but in a competitive situation made the optimal strategy doing the opposite of that heuristic oh just a few notes there's so many great points one is the game theory elferol bar problem where there's a bar and you want to go if it's under half full but not over half full it's not fun you can't get a drink and it turns out that that's like an unwinnable game because if you go well if it was busy yesterday or if it was busy three nights ago all decision rules if you just hold to them get adapted around and i think then that's a very interesting light to reject environmental adaptation as a strategy position of the book because i think adapting to the environment sounds like something biological it sounds like something that would be successful organizationally and then i think that will also lead us to some discussions of how does the free energy minimization imperative transcend the default alive expected value dialectic cool $u(m)$ vuka all right vuka so this is an acronym for volatility uncertainty complexity uh and ambiguity so i have some background noise right now by the way $u(m)$ so the i'm just reading a quote this is

actually a full disclosure from my aunt who's a leadership and organizational design consultant um she wrote this post called the truck bigs explorers in vuka territory and i really liked how she framed this where you know most of us in the west were trained in the professional model that prizes expertise today being an expert and a specialist is not enough and worse it can make you part of the problem because today we are dealing with the frontiers of humanity a place where no one has been and no one could possibly be a content expert and so what i really really like about this is that like we're entering this world that is increasingly volatile increasingly ambiguous but at the same time that really runs up against like how we're individually motivated uh as professionals where a lot of us really want to get good at a thing and have mastery and it really like requires handling this uncertainty complexity that volatility is like a personal discipline as well as an organizational discipline as far as like that personal resilience handling that uncertainty and complexity is what is going to make the organization uh resilient and that's actually quite hard to build and it's a very rare rare skill set um and a lot of it's about you know well one and i hope we like how this this graph uh frame this of like how well can you predict your actions and how much do you know about the situation and like on both of those axes uh it's becoming increasingly so that you know very little about the situation you really can't predict your actions and like the ability for uh for expertise to really navigate through that is difficult is not it's much less effective and it's much more about cross-functional um collaboration daniel blue did you want to take anything on that cool okay i'm going to read this claim from the book generative processes of top-down bottom-up hypothesis testing i.e prediction error minimization rather than top-down rule-based simplification are guiding the path to navigating a discontinuous world i'm just going to read these yellow parts in fact rather tame uncertainty in fact heuristics rather tame uncertainty heuristics reinforce dominant logics in the form of pattern recognition and tend to disregard dissonant bottom-up stimuli as noise in fact heuristics may reinforce the phenomenon of deliberate ignorance and tame curiosity's information enriching drive as a foundation for prediction error minimization um i i really liked this quote from the book and thought that it kind of um maybe sums up like the argument against heuristics and also this um top-down bottom-up circular causality that we've been discussing through the chapter daniel a little bit uncertain about heuristics for information enrichment and epistemic drive can that be a heuristic if so with that close the loop so that's like the reward function i mean we've had this this discussion many times like you know the epistemic and pragmatic value when you knock out one side of the equation what do you have left and um so yeah can your heuristic merely be info sampling i think that that's probably fair but yeah a subtle technical detail probably uh tyler cognitive maps all right cognitive maps are closely related to heuristics and can be considered the point form of visualizing sense making in form of graphic representations that locate people in relation to their information environments uh maps provide a frame of reference for what is known and what is believed um so you know i've actually made a a lot of cognitive maps myself on various consulting projects and what's kind of funny about cognitive maps is that they're actually really useful to make as like the person who is making them like as you go through the process of making a cognitive map because

you feel like oh like i understand the complexity of this organization um pretty well at this point and i can actually like hold it all in my mind at one time uh but the problem with that is that oh there's two problems one is that no one has understands your cognitive map at all it's only useful to you uh and then the other problem is that it it basically immediately becomes outdated and does not reflect how the organization actually actually works uh and so even like a month two months later it's like completely useless even to yourself um and you know john has a good quote of this where he really says that this cognitive map isn't really useful for the purposes of free energy governance so he says cognitive mapping is representative of the deterministic view of the world akin to a physical cause and effect chain and she should be considered rather anti-clocked in the face of discontinuous market environments um so yeah it's really not something that you're going to be able to fully map out um how governance works or how an organization works um this is actually something that i'm thinking a lot about for the past couple months in the context of dows of like can you model the flow of value and power completely within um an organization as code uh and i think what john would say is probably not if you can't it's not fully like this deterministic uh model of cause and effects is really something more of a social process that's a little bit more ambiguous and fluffy and and hard to nail down um daniel blue you have anything to add here

so like on the one hand i don't know like cognitive maps like when i was taught about them in college they made no sense to me i hated them i was like this is the dumbest thing i've ever heard of but then like when you get into process flow like in writing code and in like system design they kind of like make sense like like a experimental design chart so i don't know i'm back and forth on the can you read your own cognitive map sometimes like if you can't i would argue that if you can't read your own cognitive map that it's not a good one how about that uh okay um claim okay it is neuroscience's predictive coding that allows us to take management researches sense making perspective as well as nexus of structure and cognition one step further to provide a more complete and relevant framework for applying sensing and sense making in the context of strategic renewal effectively sensing and sense making are empowered by the circular causality between top-down knowledge structures prediction models predictions hypotheses and guesses and bottom-up stimuli data stemming from resource markets i just thought it was important to include this claim here i feel like it sums up a lot of what is written in this chapter um okay so we'll talk a little bit about abductive reasoning um and this picture on the left shows like inductive reasoning which is like when you have a specific observation you come to a general conclusion about um that specific observation uh like if you see you know three men wearing white shirts in a room and then you might say like oh all the men wear white shirts but it might just be a coincidence right um and then like deductive reasoning um where you have a general rule and you come to a specific conclusion like you know all the men have to be wearing white shirts in this situation and so you conclude that the men in this room will be wearing white shirts so that's like a a deductive reasoning and then you have this abductive reasoning like where you have this incomplete observation where you see like some of the men in the room wearing white shirts and you don't know anything about the rules and so so like you have this incomplete observation and you make your best prediction like um like oh i

guess some percentage of the men here like to wear white or something like that so this incomplete observation coming to this best prediction i probably did a horrible job at specific examples but those were just like sorry off the top of my head um so that's like this this reasoning logic and a quote from the text abductive reasoning usually begins with a surprising observation or experience this is what shatters our habit and motivates us to create a hypothesis that might resolve the anomaly abduction is an inferential procedure in which we create a conjecture that if it were would make part of would make the surprising anomaly part of our normal understanding of the world so i liked this idea of linking abduction to surprise and i just wanted to point out the um the live stream number 47 about active inference and abduction um and we discussed more than one paper in this live stream so just um to read a quick quote from this specific paper um this paper it's uh p pia taranin i probably butchered that and majid benny and i think majid was on that um stream with us and this paper says uh the focus is on the allegiance between the nomenclatures of active and abductive inferences as the proper place to begin reaching at the wider target the paper outlines key conceptual elements involved in a naturalistic rendering of pierce's late semiotic and logical notion of abductive reasoning and targeting a cognitive biological model of abduction which preserves the functional integrity of an organism and fulfills existential imperative for living beings evidence of existence and this paper was awesome like totally mind-blowing and i can't do it justice in like you know the five minutes or whatever time we have here so if you want to learn more about abduction and the relationship to active inference i suggest go check out this live stream or read this paper um unless you guys have comments here you can move on forward yes one comment on that is that like i found it really challenging often to like in an organizational context to toggle between inductive deductive and abductive reasoning where it's like a lot of times people get confused which kind of reasoning you're actually using if you do something that is kind of more abductive like hey like here's what i'm seeing here's my best guess like throwing things at the wall like then you get trapped into these things like well how do you really know that's that's true and you're like yeah like that's the point like i don't like i don't know that's true but like we're like really just trying to like this is a creative act we're doing together of like trying to come up with some hypothesis um i'm just like this is a really specialized skill set and it's like quite hard so yeah and funny in the book i don't think we pulled this quote out i almost did but i was like i'll leave it be but like bajon i don't know if he's quoting someone else or he just says like you know abductive reasoning it's like any hypothesis is good enough like whatever like like so it's just kind of like you know i don't know shooting arrows in the dark um it kind of made it sound like that but but there actually can be quite a bit of art and science to abductive reasoning okay cool so john lays out a few different frameworks for thinking about collective sense making and specifically like how you connect different levels of the hierarchy together like sense making at one at the individual and like you know leveling that up through organizational sense making um and so he talks about these like six different models of collective sense making and he wasn't necessarily saying like hey i am fully behind each one of these but he's more just talking about like prior work and um how to think about like the gaps and how they line up to free energy

governance so let's go through each one of them all right the four i model um so the four i models this is um the intuiting interpreting integrating and institutionalizing and so this is really thinking about how sense making flows through the individual level and then bubbles up through different levels of socialization up into institutionalization uh so crossings for i model intuiting interpreting integrating and institutionalizing as applied to the context of strategic renewal provides a framework for the process of transmorphing individual sensing and sense making cognition to firm level strategic action intuiting and interpreting a hypothesis starts at the individual level and subsequently the interpretation sense making process becomes socialized integrated and institutionalized uh at the firm level uh and so this whole process is uh is goal directed it's not just taking you know bottoms up stimuli and then somehow magically transforming it into um institutional processes it is really goal directed and hypothesis driven of uh really inactive and saying hey this is what we believe about the world and seeing is actually is actually the case um and i think john has an interesting point about the score i model that like a lot of times organizations really don't view themselves as being active creators of their uh of their world are really um especially when you contact stuff like the board views themselves as more

passive receptacles of information um and so this whole process of sense making from the individual level to the organizational level is really like an active process driven especially driven by the board blue or daniel do you have anything to add here um all right tension-based view so tension-based view this is another model that is similarly connecting both the individual level of cognition to the institutional level it just has like a different spin on it and different uh emphasis and so in this model the idea is that you need attention to make decisions right like you need to put some focus and energy to make a decision but individuals and consequently firms have limited attention right so like this is just like this fundamental constraint on your ability to make decisions is that you just don't have the time your energy to devote to every single decision um and so you know your individual cognition is constrained and because your individual cognition is constrained uh so your social cognition and like that social cognition is shaped by different factors that is like social factors cultural factors economic factors and that social cognition those forces that shape like situational context that shape your social social cognition those manifest and are institutionalized in the organizational competition and that is like rules the positions the players and the resources so if you guys are familiar with like Eleanor Ostrom's model of like thinking about um kind of how common resource organizations develop it's kind of like the same exact framework that she uses um so this is really this is very consistent with free energy

governance though it doesn't really capture like that inactive process of uh reality construction it's much more about just like thinking about attention and how that bubbles up through different levels of the hierarchy uh blue daniel anything out here all right dynamic capability view so uh the aim of dynamic capabilities research is ambitious to understand how firms can't sustain a competitive advantage by responding to and creating environmental change this is from what from help at who as one of the main proponents of this type of um of dynamic capability view uh they also go on

to say that dynamic capabilities are the uh the capacity of the organization to purposefully create extend and modify uh it's resource based uh so this is like very consistent with like how i learned about uh organizational design and evolution in like a business school is much more about like hey we have these kinds of resources kind of like a settlers of katan model almost like we have these resources and these constraints in our environment and this is like how we can extend that resource base and grow over time uh and it has a little more like a deterministic bent to it um and it's about like how you survive in a quickly changing environment so in that sense it's like similar to free energy governance uh but it's really more about like strategic change it's not really about like the individual cognition like it doesn't really talk much about individual cognition um at all so it's like missing it's like an interesting like theoretical framework but it's not really uh as embodied as free energy governance and as like um it's missing that relationship to the individual uh blue daniel and you can add here all right okay ours um so okay ours are objectives and key results so like the idea here is that you have a main so every single level of the hierarchy so let's say that your company top level of the hierarchy is like i should just use this example on the slide you want to you're the head of the brand you want to get a thousand new signups right that is your um your objective or your objective is increasing brain activations and the key result like the quantitative way you measure that is get a thousand new signups right so that's at the top level of the organization and then in each level of the hierarchy below that you have like the sub components that would need to be true for to get a thousand new signups right so the things that would need to be true to get a thousand new science are that uh well one you have to on the head of content you have to get your click-through rate above some threshold implement 10 12 new mediums post old or new content on the partnership side you have to get 10 000 event registrants uh by someday right i'm sorry man um so it's basically a way of thinking about like how goals cascade from the top level of the organization and filter down through the other levels like jesus christ sorry about that very persistent um and so this is actually really popular in a like especially like tech world uh this is pioneered by like john doer and larry page wrote this book um about measure called measure what matters it's about like a quantitative way of thinking about organizational design and goal setting and so i've actually done this a lot at different organizations and it can be powerful and i think john really likes it he says okay ours have proven to be an invaluable and dynamic tool to foster informal exchanges at the field level of resource markets approximating self-organization because each level of the hierarchy is has the responsibility for themselves to set their own eight okr that aligns with the next level of the hierarchy right so it can be like really dynamic that and uh have that self-organizing quality however a problem with it and this is not something that john brought up is something from my own experience is that like especially in a startup context things are moving so quickly that okrs very very often become outdated pretty much like immediately and so you'll set an okr for a quarter and then you're a month down the line and so much has changed like the context is so different that like like this okr obviously doesn't matter anymore um and that's a huge problem and so like you need some way of like changing your krs dynamically but the problem is is that okrs take a lot of work to set up and to socialize and to measure and

so then you like you really don't have the resources necessarily to do that and that is like a huge challenge uh there's also like huge issues with doing okrs and that it's like some uh some version of like good arts law where like when you measure when your measure becomes a target it ceases to be a good measure whereas like these different levels of the organization can over optimize for some like arbitrary metric that they need to get a promotion which sometimes that can be aligned with the organization but like in quickly changing environments it can become quickly not aligned to what the organization actually needs so there's like a lot of pushback for okrs right now it's unclear like i think there there's probably some formulation of okrs that's more dynamic and more responsive to bottom-up stimuli um but they they can be really hard to operationalize but theoretically amazing way of connecting different levels of the organization jesus christ all right different levels of the organization uh all right let's go to the next one collaborative intelligence all right so the drive for self-organizing mechanisms gradually from employee empowerment and collective intelligence in form of crowdsourcing deploying the journey the power of information technologies to tap into the collective knowledge base to call to collaborative intelligence cq emphasizing the need for a shift in the leadership model from having all the answers to raising the right questions as digital tools are increasingly deployed to enable goal-directed intelligence deliberations and distributed multi-intelligence systems and so like to to think about like what this actually means like in practice oftentimes it's not the case where it's like you have some like really huge robust system where every person in the organization is like seamlessly sharing like their insights and their data like that's not really how it works it much more works that it's like you have specialized sense makers within the organization and uh consistent ways of processing data and sharing that data so for example you'll have like uh an it team they create um they have like a central data warehouse where they put all the data together they clean it and sanitize it in a consistent way and then you have embedded kind of sense makers like kind of consultants or analysts at different levels of the organization processing that data sharing insights with each other socializing across the organization and that tends to be kind of like how it's generally structured especially in like a lot of tech companies all right an open strategy would be is kind of an extension of that it's like the extreme extension of collaborative intelligence um so open strategy is categorically different traditional top level strategizing and quant and qualitatively different from bottom-up strategizing so open strategy is unique in its overall transparency wide inclusiveness and central use of social i.t to effectively enable mass participation in summary uh open strategy is rather is rather project centric rather than a coherent new strategy governance framework the very limitations transpiring through the various case studies suggest that open strategy is likely to battle with shawarmer's uh three enemies or inner voices of resistance uh the voice of judgment the voice of cynicism and the voice of fear so like open strategy is this kind of idea where it's like a the strategy can emerge through this kind of like bottoms up information processing if we all had the information available to us wouldn't be purely on top-down decision maker for like synthesizing all of this but rather we could all do that together and something more robust will emerge through that process uh and this is like especially a popular concept among like keel organizations or holacracy that that kind of decision making

is like the gold standard for how it should happen um though like he brought some interesting use cases about uh things like ibm i think it's pepsi a few other different use cases where their experience with this is mixed and when he says this is a project centric centric rather than a coherent new strategy governance framework like it's easy to do this more in a project setting because you can think about what the relevant pieces of information are and who all the stakeholders are are that you want to involve uh but to do that dynamically means that you're like continuously bringing in new information changing a new stakeholders finding new decision making processes between those stakeholders that's super hard to do uh like on a continuous basis rather than like a limited scope project basis um so it's like a kind of unclear how to make open strategy work over like a long and definite period of time um yeah and that blue danielan thing to add here yeah i mean just like one one last point here is that like i really like how brought the voice of the inner voices of resistance the voice of judgment the voice of cynicism and the voice of fear and that's like really been kind of a this feeling i've had as i've been reading this book of like yes like this way of integrating top-down and bonds of stimuli together like sounds great and it's like totally the right way to do it but sometimes the reality of being an organization is that you're dealing with people's like personal fear and that really holds you back from being able to um to like dynamically change like people want stability people want mastery um people want things to be the word the way that they were the same in the past and like that's like a very deeply i think human thing i'm not sure if that's something that's you know unique about modern organizations uh and like you're really going up against that when you're trying to develop these more like dynamic systems of thinking about purpose and meaning and sense making um and to me in my mind that feels like the central challenge of free energy governance i'm not sure john would agree with that but that to me that feels like the huge hurdle you have to overcome cool so moving into the second section which is um much smaller so we'll try to be fast uh okay there's a claim here strategic renewal is an iterative process of belief action and learning where periods of incremental adjustment single loop learning are broken by bursts of revolutionary change double loop learning it is only when we start expressing beliefs as probability distributions and action as a state that precedes perception conditioned by a markovian boundary intermediating the internal and external states all dedicated all dedicated in circular causality to continuously optimizing the updating of beliefs i.e learning a mathematically self-consistent model emerges um so strategic renewal uh is he is using beyond is using a definition put forward in this article uh by schmidt in 2018. this book adopts a cross-hierarchical capabilities driven and action action-centric learning perspective in defining strategic renewal which is best captured by schmidt strategic renewal involves a transformation of the firm's core capabilities associated with competitive advantage concerns the entire organization and has implications across organizational levels and is essential to break path dependence and ensure the firm's long-term survival um and another claim he makes here is to approach strategic renewal as an upper echelon centric top-down only challenge is nonsense given the importance of social interactions to knowledge development and organizational learning we believe that renewal can be understood best as a system of relational or

social exchanges which is effectively the essence of reality construction indeed once strategic renewal is embraced as a system of relational or social exchanges the need for a cross hierarchical connect emerges and i just wanted to bring up this uh figure this is like the action perception loop from thinking through other minds a variational approach to cognition and culture uh we've had this on the um live stream before i can't remember and didn't look up which live stream number it was um but but in this um paper they really uh illustrate how um this relational and social exchanges like interaction with the niche environment helps us come to a mutual consensus in a social kind of way um and so if you would like more information there i recommend this paper um okay and then bijan talks about top-down versus bottom-up

strategic renewal the more dominant a top-down directed induced rather than bottom-up inspired autonomous strategic renewal process the more likely that critical intelligence from resource markets maybe that should be from resource markets is ignored and transient and transformation bearing market signals are lost in the absence of a circular top-down bottom-up organizational learning the firm may become trapped in top leadership's dominant logics highly dependent on m a joint venture driven innovation and increasingly disconnected from the very intelligence the firm's field close to resource markets beholds eventually financial control m a and alliancing is likely to squeeze out strategic control in-house innovation making the firm potentially rather more than less vulnerable to survival risk as innovation can no longer congenia can be generated from within the firm did you guys have any comments here top down versus bottom up okay um and financial versus strategic control tyler do you want to read this one for me sure uh financial strategic control respectively core like curvilinearly to confirm performance too little or too much of one or the other is potentially entropy bound the firm is challenge to find an optimal balance between external and internal innovation to power continuous strategic renewal as a sustainable foundation for firm performance and survival indeed the question arises how does this balance financial and strategic control and whether the board of directors potentially assumes critical and unique strategic role in the process

so yeah i mean i think just something to add here too is um like a lot of times his friends build or buy buy and the the startup the traditional sarcasm of like how you might balance these two decisions of like build versus buy as that like build is we call that strategic right like these are the things that are like the core and central to your purpose this is like the core like engine of how you create value in the world um and buy are the things where and buy can mean a couple of different things it doesn't actually mean necessarily it can mean more like partnerships and plugging into the external ecosystem and so that'd be like using someone else's api for something that is not necessarily central to you like stripe payment processing right and so it's this way that you can like both have the sustainable advantage while at the same time um kind of not using your resources and your energy and your attention for something that's like not core to who you are cool thank you um okay i think this is our final claim from the book strategic renewal research is in need of a more comprehensive framework structurally resolving the hierarchical top-down bottom-up learning disconnect cognitively embracing environment as a matter of enactment and in terms of capabilities balancing financial and strategic control as well

as exploitation and exploration the board of directors should be integrated as a distinct cognitive variable into the community of interactions but more importantly the question arises whether the board possesses the right capabilities to be of strategic relevance in discontinuous market environments um i think that that's all so if you guys have any comments um if not we can move into some concluding thoughts

of the book is subtitled reframing the strategy governance conversation so i think we're getting chapter by chapter a clear review of the positioning and the contribution of free energy governance and towards the middle of the book things get

really detailed with respect to the applications of the active inference framework so far we've seen more of a introductory opening salvo and a call for need for integration above and beyond the current existing practices which i think tyler did a great job um conveying like some of the real world's strengths and weaknesses and challenges of all these different strategies

uh

yeah and just to add to that like i'm really excited for um how we moves this into an applied setting because kind of like you know combining the fox and the hedgehog model like this is something i've actually thought about a lot to little success and it's a huge problem especially thinking about things like okr how quickly they become updated like how do you create

a model of the world while simultaneously communicating like hey this model is subject to change like this is a work in progress that we're co-creating together um

um and like kind of going against people's individual resistance of not wanting change and wanting things to be a staple like i don't have a clue of how you actually really do that i'm curious i really hope that sean has a good answer for that

um awesome well we'll see you guys next week uh for chapter three uh peace and enjoy your rest of your day or night wherever you might be in your temporal existence

um